

C723

Quantitative Analysis For Business

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A regional retailer with \$3 million in revenue last year is deciding whether to expand. If the market is strong (probability 0.6), expanding earns \$200,000; if weak (0.4), it loses \$50,000. Staying small earns \$90,000 in a strong market and \$40,000 in a weak one. The CFO has 20 years of experience. Using expected monetary value, what should the retailer do?

- A. Expand; EMV = \$200,000.
- B. Stay small, because its worst case is better.
- C. Expand; EMV = \$100,000.
- D. Expand; EMV = \$75,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

2. [TRANSFER — reversed stem / different variable]

A project pays \$80,000 if it succeeds and loses \$20,000 if it fails. The alternative is a sure \$30,000. At what probability of success are the two options equal in EMV?

- A. 0.50
- B. 0.25
- C. 0.375
- D. 0.30

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

3.

A manufacturer headquartered in Ohio faces high demand (probability 0.3) or low demand (0.7). Option A pays \$100,000 if high and \$20,000 if low; option B pays \$50,000 if high and \$40,000 if low. A market research firm quotes \$20,000 for a survey. What is the most the manufacturer should pay for perfect information about demand?

- A. \$20,000
- B. \$1,000
- C. \$58,000
- D. \$14,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

4. [TRANSFER — reversed stem / different variable]

A decision's best EMV is \$70,000 and its EVPI is \$12,000. What is the expected value with perfect information?

- A.** It can never be found without the full payoff table.
- B.** \$82,000
- C.** \$70,000
- D.** \$58,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A firm can build a large plant (profit \$300,000 if the economy is good, -\$100,000 if poor), a small plant (\$150,000 good, \$20,000 poor), or nothing (\$0 either way). An analyst estimates a 60% chance of a good economy. Which choice does a pessimistic (maximin) decision-maker make?

- A.** Build nothing, since it never loses money
- B.** Small plant
- C.** Large plant
- D.** Compute EMV with the 60% estimate first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A furniture maker earns \$30 per unit of X and \$40 per unit of Y. Product Y was launched last year to strong reviews. The corners of the feasible region are (0, 0), (0, 50), (40, 30) and (60, 0). What production plan maximizes profit?

- A.** 40 X and 30 Y, for \$2,400
- B.** 0 X and 50 Y, for \$2,000
- C.** 60 X and 0 Y, for \$1,800
- D.** 20 X and 20 Y, for \$1,400

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A shop's labor constraint is $2X + 3Y \leq 240$ hours. Workers are paid \$22 an hour. The optimal solution is $X = 30$ and $Y = 50$. How much slack does the labor constraint have?

- A. \$660
- B. 210 hours
- C. 160 hours
- D. 30 hours

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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8.

A distributor sells 4,500 units a year. Each order costs \$50, holding a unit for a year costs \$5, and each unit costs \$25 to buy. The EOQ is 300 units. What is the total annual ordering and holding cost at the EOQ?

- A. \$114,000
- B. \$2,250
- C. \$1,500
- D. \$750

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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9.

A remodel has five activities. A (4 days) is followed by C (6 days). B (5 days) is followed by D (3 days). E (2 days) follows both C and D. A and B start at the same time. Activity B needs a rented crane at \$900 a day. Which activities are on the critical path?

- A. All five activities
- B. A, C and E
- C. B, D and E
- D. C only, the longest single activity

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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10. [TRANSFER — reversed stem / different variable]

In the same remodel (A-C-E = 12 days; B-D-E = 10 days), activity D is delayed by 3 days. What is the new project duration?

- A. 12 days
- B. 15 days
- C. 13 days
- D. 10 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A project must finish one day sooner. Critical activity X (handled by a subcontractor with a strong safety record) takes 6 days at \$1,000 and can be crashed to 4 days at \$1,600. Critical activity Y takes 5 days at \$800 and can be crashed to 3 days at \$1,300. Noncritical activity Z can be crashed at \$100 per day. Which activity should be crashed first?

- A. Y, at \$250 per day
- B. X, because its subcontractor is reliable
- C. Z, at \$100 per day
- D. Y, at \$500 per day

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

An activity normally takes 8 days and costs \$2,000. It can be crashed to 5 days at \$400 per day saved. What is its crash cost?

- A. \$1,200
- B. \$5,200
- C. \$800
- D. \$3,200

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A regional retailer ~~with \$3 million in revenue last year~~ is deciding whether to expand. If the market is strong (probability 0.6), expanding earns \$200,000; if weak (0.4), it loses \$50,000. Staying small earns \$90,000 in a strong market and \$40,000 in a weak one. ~~The CFO has 20 years of experience.~~ Using expected monetary value, what should the retailer do?

Target word: best option by EMV **Decides:** Expand EMV 100,000 vs. stay small 70,000.

A	Wrong stem number	The best single payoff, not the expected value.
B	Adjacent category	That is the maximin rule, not EMV.
C	Correct	$0.6 \times 200,000 + 0.4 \times (-50,000) = 100,000$; stay small = $54,000 + 16,000 = 70,000$.
D	Dropped element	Averages the two payoffs and drops the probabilities.

2. Answer A Concept 1 • transfer

A project pays \$80,000 if it succeeds and loses \$20,000 if it fails. The alternative is a sure \$30,000. At what probability of success are the two options equal in EMV?

Target word: solve for the break-even probability **Decides:** $80,000p - 20,000(1 - p) = 30,000$.

A	Correct	$100,000p - 20,000 = 30,000$, so $p = 0.50$.
B	Adjacent category	$20,000 / 80,000$ is a loss-to-gain ratio, not a break-even probability.
C	Dropped element	$30,000 / 80,000$ ignores the possible loss.
D	Wrong stem number	Reads the \$30,000 figure as a probability.

3. Answer D Concept 2

A manufacturer ~~headquartered in Ohio~~ faces high demand (probability 0.3) or low demand (0.7). Option A pays \$100,000 if high and \$20,000 if low; option B pays \$50,000 if high and \$40,000 if low. ~~A market research firm quotes \$20,000 for a survey.~~ What is the most the manufacturer should pay for perfect information about demand?

Target word: EVPI **Decides:** EV with perfect info 58,000 – best EMV 44,000.

A	Camouflage	The survey quote; it does not set the value of information.
B	Confusion set	The gap between the two options' EMVs, not the value of information.
C	Dropped element	EV with perfect information, without subtracting the best EMV.
D	Correct	EMV A = 44,000, B = 43,000. With perfect info: $0.3 \times 100,000 + 0.7 \times 40,000 = 58,000$. $58,000 - 44,000 = 14,000$.

4. Answer B Concept 2 • transfer

A decision's best EMV is \$70,000 and its EVPI is \$12,000. What is the expected value with perfect information?

Target word: solve for EV with perfect information **Decides:** $EVwPI = \text{best EMV} + EVPI$.

A	Absolute language	The definition $EVPI = EVwPI - EMV$ gives it directly.
B	Correct	$70,000 + 12,000 = 82,000$.
C	Wrong stem number	The best EMV itself.
D	Inverted operation	Subtracts EVPI instead of adding it.

5. Answer B Concept 3

A firm can build a large plant (profit \$300,000 if the economy is good, $-\$100,000$ if poor), a small plant (\$150,000 good, \$20,000 poor), or nothing (\$0 either way). ~~An analyst estimates a 60% chance of a good economy.~~ Which choice does a pessimistic (maximin) decision-maker make?

Target word: maximin choice **Decides:** Worst cases: $-100,000$, $20,000$, 0 ; the best of these is $20,000$.

A	Confusion set	Its worst case, \$0, is lower than the small plant's \$20,000.
B	Correct	Its worst case, \$20,000, is the highest of the worst cases.
C	Inverted operation	That is maximax, the optimist's choice.
D	Camouflage	Maximin ignores probabilities.

6. Answer A Concept 4

A furniture maker earns \$30 per unit of X and \$40 per unit of Y. ~~Product Y was launched last year to strong reviews.~~ The corners of the feasible region are $(0, 0)$, $(0, 50)$, $(40, 30)$ and $(60, 0)$. What production plan maximizes profit?

Target word: optimal corner **Decides:** Evaluate $30X + 40Y$ at each corner.

A	Correct	Profits: 0 , $2,000$, $2,400$, $1,800$; the highest is at $(40, 30)$.
B	Adjacent category	Makes only the higher-margin product and ignores which constraints bind.
C	Confusion set	Maximizes units, not profit.
D	Dropped element	An interior point; the optimum is always at a corner.

7. Answer D Concept 5

A shop's labor constraint is $2X + 3Y \leq 240$ hours. ~~Workers are paid \$22 an hour.~~ The optimal solution is $X = 30$ and $Y = 50$. How much slack does the labor constraint have?

Target word: labor slack **Decides:** Hours used $= 2(30) + 3(50) = 210$.

A	Camouflage	Prices the slack at the wage; slack is measured in hours.
B	Adjacent category	Hours used, not unused.
C	Dropped element	$240 - (30 + 50)$ drops the coefficients.
D	Correct	$240 - 210 = 30$.

8. Answer C Concept 6

A distributor sells 4,500 units a year. Each order costs \$50, holding a unit for a year costs \$5, ~~and each unit costs \$25 to buy.~~ The EOQ is 300 units. What is the total annual ordering and holding cost at the EOQ?

Target word: total ordering + holding cost **Decides:** Ordering = $(4,500 / 300) \times 50$; holding = $(300 / 2) \times 5$.

A	Camouflage	Adds the \$112,500 purchase cost, which the EOQ trade-off leaves out.
B	Wrong stem number	Uses 300 units instead of average inventory 150 for holding cost.
C	Correct	Ordering \$750 + holding \$750 = \$1,500; at the EOQ they are equal.
D	Dropped element	Only one of the two cost components.

9. Answer B Concept 7

A remodel has five activities. A (4 days) is followed by C (6 days). B (5 days) is followed by D (3 days). E (2 days) follows both C and D. A and B start at the same time. ~~Activity B needs a rented crane at \$900 a day.~~ Which activities are on the critical path?

Target word: critical activities **Decides:** A-C-E = 12 days; B-D-E = 10 days.

A	Absolute language	Only the longest path is critical.
B	Correct	A-C-E is the longest path at 12 days.
C	Inverted operation	The shorter path, with 2 days of slack.
D	Adjacent category	Critical status belongs to the path, not the largest activity.

10. Answer C Concept 7 • transfer

In the same remodel (A-C-E = 12 days; B-D-E = 10 days), activity D is delayed by 3 days. What is the new project duration?

Target word: effect of a noncritical delay **Decides:** B-D-E becomes 13 days, exceeding its 2 days of slack.

A	Absolute language	Assumes delays off the critical path never matter; this one exceeds the slack.
B	Dropped element	12 + 3 ignores the 2 days of slack on D's path.
C	Correct	B-D-E = 13 now exceeds A-C-E = 12.
D	Wrong stem number	The old length of D's path.

11. Answer A Concept 8

A project must finish one day sooner. Critical activity X (~~handled by a subcontractor with a strong safety record~~) takes 6 days at \$1,000 and can be crashed to 4 days at \$1,600. Critical activity Y takes 5 days at \$800 and can be crashed to 3 days at \$1,300. Noncritical activity Z can be crashed at \$100 per day. Which activity should be crashed first?

Target word: cheapest crash on the critical path **Decides:** Cost per day: X = $600 / 2 = 300$; Y = $500 / 2 = 250$.

A	Correct	Lowest cost per day among critical activities.
B	Camouflage	Reliability does not change the cost per day, and X costs \$300 per day.
C	Adjacent category	Z is not critical; crashing it does not shorten the project.
D	Dropped element	Uses the total added cost without dividing by days saved.

12. Answer D Concept 8 · transfer

An activity normally takes 8 days and costs \$2,000. It can be crashed to 5 days at \$400 per day saved. What is its crash cost?

Target word: solve for crash cost **Decides:** Normal cost + 3 days × \$400.

A	Dropped element	Only the added cost, without the normal cost.
B	Wrong stem number	Uses 8 days instead of the 3 days saved.
C	Inverted operation	Subtracts the added cost from the normal cost.
D	Correct	$\$2,000 + 3 \times \$400 = \$3,200.$

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	6	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Expected monetary value	EMV = Σ (probability $\times$ payoff) for each option; under risk, choose the highest EMV.	Averaging payoffs without weights, or picking the single best payoff.	A payoff table with probabilities for each state.
2	Expected value of perfect information	EVPI = expected value with perfect information – best EMV; it caps what any information is worth.	Reporting EV with perfect information as EVPI, or comparing EMVs of two options.	“How much should we pay for a forecast or survey?”
3	Decision criteria under uncertainty	Maximax = best of the bests (optimist); maximin = best of the worsts (pessimist); neither uses probabilities.	Pulling in probabilities, or mixing up maximin and maximax.	A named attitude (optimistic, pessimistic, conservative) and a payoff table.
4	LP optimum at a corner point	A linear objective reaches its optimum at a corner of the feasible region; evaluate the objective at each corner.	Choosing the product with the best margin alone, or an interior point.	Corner points listed with a profit function.
5	Slack on a $\leq$ constraint	Slack = available resource – resource used at the solution; a binding constraint has zero slack.	Reporting resource used, or ignoring the coefficients.	A constraint with coefficients and an optimal x and y.
6	EOQ cost balance	At the EOQ, annual ordering cost equals annual holding cost; holding cost uses average inventory $Q / 2$.	Using Q instead of $Q / 2$, or adding purchase cost to the ordering-holding total.	“Total annual inventory cost” with D, S and H.
7	Critical path and slack	The critical path is the longest path; a delay on another path matters only once it exceeds that path’s slack.	Assuming any delay extends the project, or that noncritical delays never do.	A delay on an activity off the critical path.
8	Crashing	Crash the critical activity with the lowest cost per day saved: (crash cost – normal cost) / (normal time – crash time).	Crashing a cheap noncritical activity, or comparing total rather than per-day cost.	Normal and crash times and costs for several activities.

Course scope (public description): Quantitative Analysis for Business explores various decision-making models, including expected value models, linear programming models, and inventory models. This course helps student learn to analyze data by using a variety of analytic tools and techniques to make better business decisions. In addition, it covers developing project schedules using the Critical Path Method. Other topics include calculating and evaluating formulas, measures of uncertainty, crash costs, and visual representation of decision-making models using electronic spreadsheets and graphs. This course has no prerequisites.